Welcome to the 2005 Psychology and Neuroscience In-House Convention

Thanks for attending the 31th Annual In-House Convention.

Thanks should also be extended to the In-House Convention Committee for planning and organization. The committee consists of the following individuals: Natalia Jaworska, Kris Peace, Valerie Peters, Bronwyn Vigneault-MacLean, Aimee Wong and Brad Frankland.

Friendly Reminders

*Please bring your own mug for coffee or tea!
*There will be a reception at the end of Thursday in the 2nd Floor Lounge.

Thursday, April 28, 2005
Room 4258/63, Life Sciences Centre

9:00	Richard Brown: Introduction and brief history of convention.	
9:30-10:15	Session 1. Chair: Brad Frankland	
	T2)	Michael Kiefte
	T3)	Bronwyn Vigneault-MacLean
	T4)	Sarah Boudreau
10:15-10:45	Coffee Break	
10:45-12:00	Session 2. Chair: Bronwyn Vigneault-MacLean	
	T5)	Samantha Goldwater-Adler
	T6)	Valerie Grant
	T7)	Jennifer Parker
	T8)	Dilys Leung
	T9)	Heléne Deacon
12:00-1:30	Lunch Break and Posters (Psychology 2nd Floor Lounge)	
	P1)	Matthew Murphy
	P2)	Penny Corkum
	P3)	Amélie Doucet
	P4)	Anne-Claire Larochette
	P5)	Allison MacFadyen
1:30-3:00	Session 3. Chair: Aimee Wong	
	T10)	Ray Klein
	T11)	Mike Lawrence
	T12)	Esther Lau
	T13)	Darren McKee
	T14)	Angela Arsenault
	T15)	Margo Watt
3:00-3:30	Coffee Break	
3:30-4:45	Session 4. Chair: Sophie Jacques	
	T16)	Cindy Hamon-Hill
	T17)	Lindsay Uman
	T18)	Melissa McGonnell
	T19)	Erin Moon
	T20)	Stephen Lewis
5:30	Dinner / Reception in the 2nd Floor Lounge	

Friday, April 29, 2005
Room 4263, Life Sciences Centre

9:00-10:15 Session 5. Chair: Brad Frankland
- T21) Henning Breyhan
- T22) Jennifer Stamp
- T23) Natalia Jaworska
- T24) Rahia Mashoodh
- T25) Simon Gadbois

10:15-10:45 Coffee Break

10:45-12:15 Session 6. Chair: Jennifer Stamp
- T26) Amanda Poulin
- T27) Biljana Stevanovski
- T28) Alana Paul
- T29) Dominic Standage
- T30) Kristie Dukewich
- T31) John Christie

12:15-1:30 Lunch Break and Posters (Psychology 2nd Floor Lounge)
- P6) Jonathan M. Fawcett
- P7) Autumn P. Mochinski
- P8) Nicola Hoffman

1:30-3:00 Session 7. Chair: Aimee Wong
- T32) Chris Mushquash
- T33) Sherry Stewart
- T34) Mandy Emms
- T35) Antonina Kuznetsova
- T36) Eva Gunde
- T37) Jeffery MacLeod

3:00-3:30 Coffee Break

3:30-4:30 Session 8. Chair: Bronwyn Vigneault-MacLean
- T38) Frederick Li
- T39) Steve Shaw
- T40) Yoko Ishigami
- T41) David Westwood
- T42) Joshua Salmon

4:45 Bronwyn Vigneault-MacLean - Closing Remarks

5:00 RECEPTION

T1) A historical introduction to the In-House Conference

Richard Brown

This presentation will use photographs from the past 30+ years to introduce the in-house. Photos of past faculty, students PDF's in their most embarrassing moments will be presented. Test your knowledge in the who's who of famous faces in Psychology at Dalhousie.

T2) Auditory Representation of Vowel Spectra

Michael Kiefte and Sara Macintosh

Several attempts have been made to estimate the auditory representation of vowel spectra in human listeners analogous to what has been obtained via single unit recordings in the auditory nerve of non-human animal surrogates. One such attempt with human subjects uses the pulsation-threshold technique to evaluate masking patterns with complex maskers such as steady-state vowels. However, it is not clear what this psychoacoustic task is measuring and there is a clear potential for strong bias effects. This paper describes a preliminary attempt to eliminate some of that bias and presents data from several listeners. We also discuss some remaining problems with the paradigm.

T3) “Time after Time”: Central Auditory Aftereffect for Interaural Time Difference

Bronwyn K. Vigneault-MacLean and Dennis P. Phillips

This research employs psychophysical methods to explore the neural mechanisms that the brain uses for localizing sounds in space. We are using a selective adaptation paradigm to interfere temporarily with part of the mechanism, and measuring the effect of this on a listener’s ability to localize sounds. This paradigm investigates the ability of subjects to localize tonal stimuli on the basis of interaural time differences (ITDs). The subjects report the absolute position from which they perceive the sound to come – and do so before and after exposure to an adapting stimulus that consists of tones strongly lateralized to one or other side. The adaptation results in a “flattening” of the Localization vs ITD function, i.e., the range of reported positions of the test stimulus, is narrowed after exposure to the adapting stimulus. These effects inform us about the architecture of the neural machinery for sound localization.

T4) Caffeine's role in conditioning preference and palatability shifts.

Sarah Boudreau and Vin LoLordo

Caffeine is one of the most widely consumed psychoactive substances in the human population. Human data indicate both an increased wanting and liking for substances containing caffeine. The current study sought to further examine the role of caffeine in liking and wanting. Rats were implanted with intraoral cannula that allowed infusion of solution directly into the mouth. On CS+ trials grape or cherry Kool-Aid was mixed into a tap water solution with caffeine, and on CS- trials the other flavor was mixed with water alone, in a counterbalanced procedure. Rats were given thirteen days of training; six days with the CS+ solution, and seven days with the CS- solution. Experimental bottles were placed on the cages overnight, and removed the next morning in a single alternation schedule. Following training, rats were given a taste reactivity test (TRT) with each Kool-Aid-water solution. Each TRT consisted of placing the rat in the TR chamber, and infusing the Kool-Aid solution for 2 minutes while recording the rat's orofacial reactions. Immediately following the TRTs, all rats were given two overnight two-bottle preference tests. Both groups showed a preference for the CS+ in the two-bottle test, but there was no evidence from the TR data of an appetitive palatability shift towards the CS+. These data indicate that while the rats learned to want the Kool-Aid previously paired with caffeine more then the Kool-Aid paired with water, they did not learn to like it any more then the CS-.

T5) Bullying of Children with Autistic Spectrum Disorders: A Review of the Literature

Samantha Goldwater-Adler and Susan E. Bryson

To date, the empirical literature on the prevalence and nature of bullying of youth with autistic spectrum disorders (ASD) is sparse, at best. However, clinical, parental and first-hand accounts all implicate an increased prevalence of peer-victimization and social exclusion for this group as compared to that for typical children (e.g., Little, 2002; Jackson, 2002), consistent with reports for children with other learning and intellectual disabilities (e.g., Mishna, 2003). Furthermore, the key characteristics of victims of bullying correspond to the primary deficits found in those with ASDs, specifically, poor social skills, few friendships (Griffin & Gross, 2004; Yude et al., 1998), and negative communication beliefs and abilities (Storch et al., 2002). The severity of the adverse outcomes of bullying, including increased risk of mood and anxiety disorders (Griffin & Gross, 2004) and even suicide (Coggan et al., 2003), is an issue of pressing concern. Our review of the relevant literature focuses on recommendations for future research and for the development of intervention strategies oriented to the special needs of children with ASDs and related learning problems.

T6) Impact of Positive & Anxious Mood on Alcohol-Related Cognitions in Internally Motivated Drinkers

Valerie V. Grant and Sherry H. Stewart

This study investigated the impact of mood on the alcohol-related cognitions of undergraduates who drink to cope with anxiety (CM-anxiety) versus those who drink to enhance positive mood states (EM). Participants were randomly assigned to either a positive or anxious musical mood induction condition. We expected that EM (but not CM-anxiety) participants would exhibit increases in explicit emotional reward expectancies following exposure to the positive mood induction procedure and that CM-anxiety (but not EM) participants would show increases in explicit emotional relief expectancies after exposure to the anxious mood induction procedure. Also, we anticipated that EM (but not CM-anxiety) participants would have longer colour-naming latencies (in a Stroop task) for alcohol versus non-alcohol targets in the positive mood condition and that CM-anxiety (but not EM) participants would have longer colour-naming latencies for alcohol versus non-alcohol targets in the anxious mood condition. Preliminary results will be presented.

T7) Lions and Tigers and Bears – Oh My! Conceptual Categorization in Adults.

Jennifer Parker & Sophie Jacques

Conceptual categorization, grouping objects into structured categories, is a critical developmental achievement. People can group objects thematically (objects co-occurring in the same event; lion and circus), perceptually (similar looking objects; ball and apple), or taxonomically (objects belonging to the same semantic category; tiger and bear). Preschoolers have thematic and perceptual biases, whereas older children tend to match taxonomically. However, in exp. 1 with adults, we found that they also show a first-choice selection bias for matching thematically. In exp. 2, we examined the role of language in adults' categorization preferences using a dual-task paradigm. Participants were given 3 versions of a match-to-sample task in which they selected two objects that were related to the target object in different ways while either doing nothing, saying a word repeatedly, or smacking their lips. Results suggest that language may be important for categorization as participants took longer to make their first selection in the verbal-interference condition than in the other conditions.

T8) How do kids spell?

Dilys Leung and Heléne Deacon

The way children think about words may have an influence on how they spell. Many studies have shown that young children spell based on how words sound (e.g. Read, 1986). However, other research has shown that young children use their knowledge of morphemes, or the smallest units of meaning, to help them spell (e.g. Treiman & Bourassa, 2000). The current study investigates young children's use of morphological information when spelling inflections and derivations. Children are asked to choose the spelling for the same word-final sound sequence in 3 types of words: inflected (–er; e.g. smarter); derived (–er and –or; e.g. teacher and director); and one-morpheme (–er and –or; e.g. number and author). Preliminary results indicate that 6- and 7-year-olds have some knowledge of the relationship between morphemes and spelling. The results will be discussed in the context of current theories of spelling development.

T9) Criss-cross: The role of language-skills in reading in developing bilinguals

Heléne Deacon

We report on the results of a three year study following six-year-old English speaking children in a French immersion programme. This study focused on the roles of language-related skills, such as phonological and morphological awareness, in reading development. Results show that skills in each language contribute to reading in the other, showing that skills can be used across languages. Intriguingly, the patterns of contributions change as children's French language and reading abilities develop. Results will be discussed in the context of current theories of bilingual reading development.

T10) If, "Everyone knows what attention is..." then why should you come to this talk?

Raymond Klein

To learn: What are the components of attention and How to measure them. Attention, the gateway to awareness and memory and according to Titchener the conceptual fulcrum for psychological science, is not a single concept with a simple, underlying machinery. Rather attention refers to a complex, interrelated set of components that can be construed as isolable subsystems or as separable, interconnected networks. Posner & colleagues have designed a simple test, the Attention Network Test (ANT), for assessing the efficacy of three of the components of attention: alertness, orienting and executive control. Scholars are excited by the scientific progress that might follow from the conceptual distinctions implicit in this view of attention and by the simple test battery for assessing these three isolable components of attention. I will critically review this test, describe some studies that have used it and highlight an improved version recently developed in Spain.

T11) The Median is Not the Message: Ex-Gaussian Decomposition of Reaction Time Distributions and Application to Measures of Attention

Mike Lawrence and John Christie

Reaction time (RT) distributions are typically positively skewed, indicating that conventional analyses using Gaussian statistics (mean, median, etc) may inaccurately represent data relative to more informative distribution models. One alternative model, the Ex-Gaussian distribution, provides three descriptive parameters: Mu (a measure of central tendency), Sigma (a measure of variance) and Tau (a measure of skewness). Estimation of these parameters is called Ex-Gaussian Decomposition (EGD). In this talk, a brief review of past psychological research using EGD on RT data will be presented, followed by discussion of the technical details of applying EGD to RT tests of attention. A Monte Carlo simulation will be reviewed that tests and corrects a popular EGD algorithm so that it can be validly applied in situations where the number of observations per participant is limited.

T12) Validating and Modifying a Healthy Model of Visuo-Spatial Neglect.

Esther Y. Y. Lau, and Gail A. Eskes

Visuo-spatial neglect is characterized by an impaired ability to orient, detect and/or respond to stimuli on the opposite side of the lesioned hemisphere. Studies of visual orienting ability using a cuing paradigm suggest neglect is related to a disengage deficit (DD) as measured by significantly increased costs when targets appearing on the poor side are preceded by an invalid cue. Although neglect is common after right hemisphere stroke, the heterogeneity of neglect presentation renders large-scale, randomized-trial study on the phenomenon and its rehabilitation strategies extremely difficult. The present study aimed at validating and extending a model of the disengage deficit in visuo-spatial neglect in healthy individuals as recently reported by Vecera and Flevaris (in press). Twenty-four normal young adults were tested using a covert orienting target detection task with exogenous cueing at 3 stimulus-onset asynchronies (SOAs). Unlike Vecera & Flevaris, but similar to neglect patients, our results showed a DD pattern only when the left side was degraded. Furthermore, and consistent with previous studies with neglect patients, the DD pattern decreased with increase in SOAs. Development of this model may be a useful method to explore some aspects of visuo-spatial neglect in normal people

T13) Reflexive orienting to gaze cues that look away from centre and then return does not produce inhibition of return.

Darren McKee, John Christie and Raymond Klein

Orienting to an uninformative peripheral cue is characterized by a brief facilitation followed by a long lasting inhibition once attention is removed from the cued location. Although central gaze cues cause automatic orienting, no one has yet seen the delayed inhibitory effect. We hypothesized that this might be seen but only if attention were effectively returned to center by a return gaze. Normal subjects were presented with a central schematic face gazing straight ahead. Following a shift in gaze, accomplished only by an apparent movement of the eyes within the schematic face, to the left or right either (i) the gaze continued in this direction, (ii) the gaze returned to centre, or (iii) the outline of the face was made bolder. The time-course of gaze-directed orienting was measured by varying the interval between the gaze cue and a peripheral target appearing randomly to the left or right of the central face. Despite our efforts to unmask inhibition of return by ensuring that attention was removed from the cued (gazed at) location we robustly failed to elicit IOR in this paradigm.

T14) Don't Ignore Me! The Role of Directed Attention in Directed Forgetting

Angela Arsenault and Tracy Taylor-Helmick

A standard item-method directed forgetting task presented sequential word pairs consisting of two differently colored words. Participants were instructed to attend or ignore designated words based on color. Immediately following the each word pair, a remember or forget memory cue was presented aurally. Succeeding study, a recognition test measured participants' retention of both forget and remember cued words, regardless of attention instruction (attend or ignore). Percentage correct scores for attended words illustrate superior recognition performance on remember-cued as compared to forget-cued words, indicative of a robust directed forgetting effect. In contrast, this effect was not evident from recognition performance for ignored words. Interesting trends, such as the potential for a reverse directed forgetting effect for ignored words, and the role of fixed attention capacity, will be discussed.

T15) Music: Cheescake or a primer of social cognitions?

Cindy Hamon-Hill and Bradley Frankland

Music has been identified as integral to human emotional and social development and may play a strong functional role in strengthening alliances between like-minded individuals. It is proposed that music may have a priming effect on mechanisms underlying social cognitions. An outline for a study investigating a priming effect by a musical manipulation will be presented. Subjects exposed to a musical treatment will be asked to interpret short video segments of animated geometric shapes. The quality and quantity of social attributions in their responses will be compared to those of a control group. Methodology and expectations will be discussed.

T16) A Brief Cognitive-Behavioral Treatment Designed to Reduce Anxiety Sensitivity: Evidence from a Randomized Controlled Trial.

Margo C. Watt and Sherry H. Stewart

Anxiety sensitivity (AS) refers to the fear of arousal-related sensations. AS is a known risk factor for anxiety and related disorders (e.g., depression, substance abuse). For the past two years, a brief cognitive behavioral (CBT) group intervention has been conducted with first year undergraduate female students at Dalhousie and StFX Universities. Participants were classified as either high or low in AS, according to their scores on the Anxiety Sensitivity Index (ASI; Peterson & Reiss, 1992). They then were randomly assigned to participate in 3 one-hour, small group sessions of either CBT to lower AS (psychoeducation, cognitive restructuring and physical exercise interoceptive exposure) or a control group seminar (discussion about psychology ethics). Measures of anxiety, as well as depression and drinking behaviour, were assessed at three time points: pre-, immediately post-, and approximately 10 weeks post-participation. Findings are discussed in terms of implications of such treatment for the prevention of important psychological disorders.

T17) Psychological interventions for needle-related procedural pain: Results of a systematic review for the Cochrane Collaboration

Lindsay Uman, Christine Chambers, Pat McGrath and S. Kisely

The purpose of this review was to assess the efficacy of psychological approaches for the management of needle-related pain and distress in children. We conducted a systematic review for the Cochrane Collaboration of all randomized controlled trials (RCTs) examining psychological interventions for needle pain and distress in children aged 2 to 19 years. A variety of cognitive behavioural approaches were examined. Pain and distress outcomes were assessed by self and observer report, as well as observational and physiological measures. Preliminary searches yielded 137 possible studies; of these 17 met criteria and were included (N=1311). Weighted mean differences with 95% confidence intervals were calculated for all outcome categories. Preliminary analyses indicate that psychological interventions, particularly distraction, hypnosis, preparation, memory alteration, and coaching were effective in reducing pain and/or distress. Insufficient data were available to assess the efficacy of a number of other common cognitive-behavioural interventions. Preliminary results provide support for the efficacy of certain psychological interventions for reducing needle pain and distress.

T18) Programme Evaluation of the Colchester East Hants Attention Deficit Hyperactivity Disorder Clinic

Melissa McGonnell and Penny Corkum

The purpose of this study was to conduct a programme evaluation of the Colchester East Hants Attention Deficit Hyperactivity Disorder Clinic (ADHD Clinic). The ADHD Clinic is a partnership between the Colchester East Hants Health Authority and the Chignecto Central School Board with strong ties to Dalhousie University and has been operating since the fall of 2000. Informal feedback has led to the demand for increase and expanded services by the ADHD Clinic. Before undertaking any enhancement of the services it currently offers, it was decided that an evaluation of current services as well as the satisfaction of the stakeholders in the ADHD Clinic and of its clients should be undertaken. In Phase 1 of this evaluation, stakeholders were interviewed regarding their thoughts about their roles as well as the current functioning of the clinic with regard to teamwork, the organisation of the clinic day, and the information provided to families and schools. Stakeholders had generally very positive feelings about the ADHD Clinic along with many suggestions for future directions. Stakeholders were also asked about their thoughts about Phase 2 of the evaluation. In Phase 2, surveys will be mailed to parents, family doctors, and teachers, of the children who have attended the clinic to date, so as to determine their beliefs about the services provided by the ADHD Clinic and their suggestions for possible improvements.

T19) Painful procedures in the pediatric emergency department: A DVD-based intervention program for child pain and anxiety

Erin C. Moon, Patrick J. McGrath, Christine T. Chambers, Douglas Sinclair and Eleanor Fitzpatrick

This talk will provide an overview of a DVD-based intervention program developed for school-aged children undergoing painful procedures in the emergency department. The intervention program is designed to reduce the pain and anxiety children experience when they have venepunctures (needles) as part of their emergency department care. Two short DVD's are included in the intervention program, both of which will be presented to children and their parents on a portable DVD player in the emergency department. The first DVD instructs children and parents on how to use distraction and relaxation techniques, which recent studies have shown to be effective at reducing children's pain and distress during medical procedures. The second DVD is an age-appropriate movie that will be used to distract children from their venepunctures. In addition to the overview of the intervention program, this talk will describe a research protocol developed in order to examine the effectiveness of the intervention.

T20) The development of a measure to elucidate motivations for self-injury: The Self-injury Motivations Scale (SIMS)

Stephen Lewis

To date, empirical and theoretical works have attempted to articulate motivations for self-injury (SI). The research conducted thus far has examined SI-motivations in the context of specific psychiatric illnesses (e.g. borderline personality disorder), in the context of self-poisoning only, and without accounting for proximal and distal concomitants of SI. Though useful, these finding are limited in terms of generalizability and clinical utility. Furthermore, there is currently no psychometrically sound measure by which to effectively assess motivations for SI. As such, the purpose of the present study was two-fold. First, a scale was developed to examine motivations for SI and psychometric properties were computed. Second, motivations for SI, extracted from the newly developed measure, were examined in the context of proximal and distal psychosocial SI-correlates (e.g., sex, history of abuse, sad mood). Results, clinical implications and directions for future research will be discussed.

T21) Modelling Alzheimer's Disease in transgenic mice

Henning Breyhan and Thomas A. Bayer

The 'amyloid hypothesis' proposes that ß-amyloid is the cause of Alzheimer's disease pathogenesis. Although many transgenic mouse lines develop ß-amyloid plaques in the brain, the link between amyloid plaques and neuron loss has not clearly been established. We identified substantial age-related neuron loss in the hippocampal pyramidal cell layer of APP751/PS1 double-transgenic mice at 16 months of age. These APP751/PS1KI mice produce abundant Aß42 with an advanced onset of plaque formation, which correlated with the PS-1 knock-in gene dosage. Six month old mice hemizygous for APP751 and homozygous for PS1KI revealed a dramatic hippocampal neuron loss predominantly in CA1. Thus the APP751/PS1KI mouse model shows major neuropathological hallmarks with abundant neuron loss that is clearly independent from extracellular amyloid deposition and identifies intraneuronal Aß as a major neurotoxic risk factor. We will now conduct behavioural tests on these mice.

T22) Individual differences in locomotor reactivity to novelty predicts expression of the transcription factor delta FosB in the central reward circuitry of the mouse

Jennifer Stamp

In laboratory rodents, individual vulnerability for self-administration can be predicted by locomotor response to a novel environment. Animals that show high behavioural reactivity to a novel environment (known as high responders or HR) acquire self-administration much more readily than animals that display lower behavioural reactivity (low responders or LR). The present study attempted to characterize molecular differences between HR and LR in the nucleus accumbens and striatum of mice. C57 mice were first identified HR or LR in an open field test; subsequently, Western blotting was used to determine if differences also exist in gene expression. Levels of the transcription factor, delta FosB, were found to be significantly enhanced in the nucleus accumbens of HR compared to LR. The downstream protein cyclin-dependent kinase 5 (Cdk5) did not differ between the two groups, neither in the nucleus accumbens nor dorsal striatum. Both delta FosB and Cdk5 are up-regulated after chronic drug exposure and are thought to mediate changes in the brain involved in addiction. Our findings that individual differences exist in delta FosB expression at basal levels in the reward circuitry of the brain may further clarify the long-term neural and behavioural plasticity changes involved in addiction.

T23) Behavioural and neurochemical effects of early stress and Substance P antagonists in a gerbil model of human depression

Natalia Jaworska, Suzanne Dwyer and Benjamin Rusak

The proposed research will assess an animal model of human depression, the maternal separation paradigm, using a species infrequently studied in relation to depression. The reason for using this species (Mongolian gerbil; *Meriones unguiculatus*) relates to the potential role of the neuropeptide Substance P (SP) in depression and/or anxiety. Several pre-clinical and clinical studies suggest that antagonism of SP receptors (neurokinin [NK]-1R), which are structurally very similar in humans and gerbils, but not in most other rodents, results in behavioural changes similar to those produced by treatment with standard antidepressants, but with fewer side effects. We will also examine the neurochemical and behavioural changes induced by early maternal separation in adult gerbils of both sexes. The lack of studies using depression and anxiety tests in gerbils also necessitates evaluating the appropriateness for this species of various behavioural tests commonly used for rats and mice.

T24) Smell Hath This Furry Woman Warned: The Effect of Predator Cues on Maternal Behaviour and the Implications for Offspring Development

Rahia Mashoodh, Jane McLeod, Alexandra Karas, and Tara Perrot-Sinal

In rodent mothers, variations in maternal behaviour during the first week of life are associated with the development of the stress response in offspring. We hypothesized that plasticity of maternal care in rats may exist to serve as the interface between the offspring and environmental conditions, such that parents can alter both physiology and behaviour of adult offspring in accordance with present environmental demands. We have started investigating this hypothesis using experiments in which rat mothers are exposed to the threat of predation after giving birth (GB). We have demonstrated that acute (1hr; given 5-6h GB) and chronic (1hr/day for 10d GB) exposure to cat odour has lasting effects on particular maternal behaviours exhibited by the rat mother. Preliminary work suggests that the offspring raised by predator-exposed mothers are generally less anxious but maintain a robust behavioural response to cat odour exposure in adulthood. Taken together, it is our contention that the use of more ecologically relevant stimuli may be more suited to the study of environmental impact upon the programming of the developing brain.

T25) Behavioural ecotoxicology: Why you should care about the mummichog

Simon Gadbois and Shannon Bard

Behavioural ecotoxicology is a field in full expansion. Data is accumulating suggesting that many environmental neurotoxins and endocrine disruptors are changing the natural action patterns of many vertebrate species, influencing reproduction, foraging, homing, predator avoidance, social interactions, etc. We trace the history of this new field and present conceptual issues emanating from the convergence of many different areas (behavioural neurosciences, ecotoxicology, psychology, ethology, behavioural ecology, etc.) working towards the understanding of neurotoxins and endocrine disruptors. We also present a common animal model in ecotoxicology, the common killifish, or mummichog (*Fundulus heteroclitus*) that is unfortunately not well documented in ethology. We are working on an exhaustive ethogram of the species as well as specific behavioural assays that will allow the detection and/or assessment of neurotoxins and endocrine disruptors on the nervous and endocrine systems.

T26) Cholinergic basal forebrain neurons containing vesicular glutamate transporter 3 project to the amygdala.

Amanda Poulin and Kazue Semba

The cholinergic basal forebrain (CBF) plays a major role in behavioural and cortical arousal. It has been suggested that some CBF neurons may co-release glutamate. A subpopulation of CBF neurons has been reported to contain vesicular glutamate transporter 3 (VGluT3). This study characterized this subpopulation of CBF neurons by examining their distribution and projections, using immunofluorescence for choline acetyltransferase (ChAT), VGluT3 and p75 nerve growth factor receptors, and retrograde tracing. ChAT+VGluT3+ neurons are located throughout the BF. Few of these neurons are positive for p75. ChAT+ axon terminals found in the neocortex, hippocampus and thalamus were devoid of VGluT3. The basolateral amygdala, which receives projections from p75-negative CBF neurons, contained numerous ChAT+VGluT+ terminals. ChAT+VGluT+ BF neurons were retrogradely labeled from the BLA. These findings indicate that a subpopulation of BF cholinergic neurons containing VGluT3 might have the capacity to release both glutamate and acetylcholine in the amygdala.

T27) Can the N2pc be observed in the absence of distractors?

Biljana Stevanovski and Pierre Jolicoeur

Previous research suggested that the N2pc, an electrophysiological component believed to reflect the allocation of visuospatial attention, reflects distractor suppression (e.g., Luck & Hillyard, 1994) and is not observed for a single target in the absence of distractors. However, others have argued that the component reflects target selection (e.g., Eimer, 1996). We examined whether the N2pc is observed in the absence of distractors. Subjects viewed a lateralised line followed 500 ms later by a centrally presented digit. A larger N2pc was observed when subjects performed a line length discrimination task than when subjects performed a digit identification task, suggesting that this component can be observed in the absence of distractors.

T28) Caffeine evoked activation in the basal forebrain: The quest for the nameless neurons

Alana Paul, E. Cumyn, Samuel Deurveilher and Kazue Semba

Caffeine is a commonly used drug that acts in the brain to increase arousal. Using c-Fos as a marker of neuronal activation, we recently showed that neurons in the substantia innominata (SI), a wake-promoting region of the basal forebrain, were activated following injection of caffeine; these neurons were neither cholinergic nor parvalbumin-positive. We examined other possible phenotypes of caffeine activated neurons using double-labelling for c-Fos or Egr-1 (another neuronal activation marker) and the calcium binding proteins, calbindin-D and calretinin. We found that a moderate dose of caffeine (30 mg/kg i.p.) increased locomotor activity; this increase was positively correlated with the number of cells immunoreactive for c-Fos, but not Egr-1, in the SI. The c-Fos or Egr-1 expressing neurons did not colocalize with neither calbindin-D nor calretinin. These results indicate that the caffeine-activated neurons in the SI are not cholinergic nor contain the calcium binding proteins examined, and so the identity of these neurons is still unknown.

T29) A computational model of divided visual attention

Dominic Standage, Thomas Trappenberg and Ray Klein

Experimental evidence on the distribution of visual attention supports the idea that environmental and cognitive influences on attention are integrated by a winner-take-all mechanism. We implement this mechanism with a computational model, testing the ability of our model to explain divergent evidence on the spatial allocation of attention. The majority of evidence supports the view that attention is unitary, but recent experiments suggest that the focus of attention may be split. We simulate two such experiments. Our results suggest that the ability to divide attention depends on sustained short-term memories, stressing the interplay between attention and working memory.

T30) Comparing target detection, target localization, and target identification in a visual search paradigm

Kristie R. Dukewich and Raymond M. Klein

Visual search is a fundamental everyday activity and one of the most frequently used tasks to study attention. The majority of search experiments require the participant to signal the presence or absence of a pre-specified target; however, everyday searching more typically entails finding a target that is present and localizing it (finding one's car in a parking lot) or determining one of its properties (does the yellow car have its lights on). Yet, studies of localization and identification are relatively rare, and we are aware no studies that have compared the three tasks. In the present study, participants searched an array of rotated Ls for a target, T, rotated either 90 deg. or 270 deg. from upright. Each participant searched using each of the three different tasks, counterbalanced for order. Participants were more efficient at finding the target in the present/absent task, compared to the localization and identification tasks, which did not differ. Moreover, slopes for localization and identification were significantly correlated, but neither of these correlated significantly with "present" slopes suggesting - despite conventional wisdom - that "finding the target" is a common stage underlying the slope in all three task - that the tasks may be eliciting fundamentally different search strategies.

T31) Is moving right to left from left of left harder than moving right to right from left of left after being hit in the right side of the head?

John Christie & Gail Eskes (in collaboration with Dave Westwood, Paddy McMullen, Ray Klein & Matthias Schmidt)

Husain et al (2000) found that neglect patients were impaired in initiating a movement to the left when the damage was more parietal than frontal. This effect was independent of whether the stimulus location was to the left or right of fixation and is believed to be a motor programming deficit only in the parietal neglect patients. We have replicated their experiments in normals and imaged the behaviour using fMRI. This presentation will focus on the methods, new performance findings, and adventures in functional imaging.

T32) The Structure of Drinking Motives in Mi'kmaq Adolescents

Christopher Mushquash, Sherry Stewart, Patrick McGrath and Nancy Comeau

When psychological measures are used with cultural groups that are different from the sample the measures were developed with, it is important to evaluate the psychometric properties within the new group. The Revised Drinking Motives Questionnaire (DMQ-R; Cooper, 1994) is a 20-item self-report measure designed to quantify adolescents' reasons for drinking alcohol. Specifically, it was designed to tap levels of the four drinking motives in Cooper's model (i.e., social, enhancement, coping, and conformity). Respondents rate their relative frequency of drinking for each of the indicated reasons on a 5-point Likert scale. The factor structure of the DMQ-R was examined in a sample of First Nations (i.e., Mi'kmaq) adolescents. Exploratory factor analysis was used because the factor structure of the DMQ-R has not been examined previously in this group. Analyses indicated a three-factor structure, with enhancement and social motives not separating into two distinct factors. Moreover, community informants (e.g., school personnel) anecdotally indicated possible problems with some of the social motive items in this cultural group. A qualitative methodology will be used to determine potential reasons for the discrepancy from previous findings in the majority culture (i.e., measurement problem vs. real difference in the structure of drinking motives in the Mi'kmaq culture).

T33) Relations between personality variables and risky drinking motives in aboriginal adolescents.

Sherry Stewart, Kaitlin English and Nancy Comeau

164 high school students from two Aboriginal communities in NS completed a measure of personality risk for substance abuse – the Substance Use Risk Profile Scale [SURPS] – which taps anxiety sensitivity (AS), hopelessness (HOP), sensation seeking (SS), and impulsivity (IMP). The 96 drinkers also completed the Drinking Motives Questionnaire – Revised, which taps coping, conformity, enhancement, and social drinking motives. AS was correlated with conformity drinking motives. HOP was marginally positively correlated with coping drinking motives and significantly negatively correlated with enhancement and social motives. SS was correlated with enhancement and social motives. IMP was correlated with all four drinking motives. The SURPS appears a valid measure for tapping personality variables related to risky drinking motives in Aboriginal teens. Similarities and differences between the present findings and those observed in a sample of teens from the majority culture will be discussed with emphasis on possible cultural factors contributing to the observed differences.

T34) MRI-based anatomical analysis of the caudate nucleus in children with attention-deficit/hyperactivity disorder.

Mandy Emms, Denise Bernier, Frank MacMaster, Benjamin Rusak & Normand Carrey

Attention-deficit/hyperactivity disorder (ADHD) is a childhood onset behavioural disorder that affects 3-5% of school age children and is characterized by inattention, impulsivity and hyperactivity. Anatomical studies using magnetic resonance imaging (MRI) have reported differences between controls and children with ADHD in the size and structure of the caudate nuclei, but results have not been consistent across studies. In this study, two observers, blinded with respect to group assignment, quantified the size and shape of the caudate nuclei of 6 boys (6-8 years of age) and 6 age-matched controls using manual tracing of the whole caudate in MR images. Analysis included detailed volumetric measurements of the caudate in each section through the nucleus, studied in the sagittal, coronal and horizontal planes. Analysis revealed that while the ADHD children did not differ in total caudate volume from controls, they showed a significant localized volume loss. The main finding of the study was that the central region of the left caudate was significantly smaller in the children with ADHD, as well as the caudal region of the left caudate and the rostral region of the right caudate. These findings of localized volume loss on both the left and right caudate nuclei help elucidate why previous neuroimaging studies have found such varied results when assessing total caudate volume.

T35) Tissue Type Classification of Brain Regions of Interest in Magnetic Resonance Images

Antonina Kuznetsova , Denise Bernier, Martin Alda, T. Hajek, Benjamin Rusak and John Fisk

Magnetic resonance spectroscopy (MRS) is a method for examining concentrations of metabolites in circumscribed areas of the living brain (regions of interest; ROIs). Any given ROI may contain different tissue types (white matter, grey matter and cerebrospinal fluid), the proportions of which will affect interpretation of MRS data. The purpose of this study is to compare the ability of two commonly used software packages (Brain Voyager QX v. 2 and AFNI) to perform automated tissue-type classification of ROIs accurately and reliably. We will use twenty anatomical (T1-weighted) MR brain images collected using a 1.5 Tesla General Electric scanner and containing 60 ROIs in the in the left medial temporal lobe, in the right and left dorsal pre-frontal cortex, and in the right and left medial pre-frontal cortex. Two experimenters familiar with these methods will independently use the standard AFNI and Brain Voyager procedures for tissue type classification on these ROIs. Intra-rater reliability of 95% and above and inter-rater reliability of 90% and above will be considered acceptable. Validity of classification results produced by each method will be assessed by a blind expert rater who will be asked to visually assess each ROI and to determine which software provided the most accurate representation of tissue type proportions in each one. The outcome measure will be the frequency with which each method is chosen.

T36) Validation of High-field MRI Volumetric Assessment of Morphological Changes in Olfactory Structures in Humans

Eva Gunde and Kim Good

Recent neuroimaging studies have reported significantly smaller than normal olfactory bulb volumes associated with impaired olfactory performance in patients with schizophrenia and related psychotic disorders. The small dimensions of the olfactory bulb and tract, however, make *in vivo* volumetric assessment of these structures difficult. The present study was undertaken to investigate the feasibility of using of high field (4T) structural MRI to perform volumetric analysis of the olfactory bulb and tract in patients with psychotic disorders in comparison with healthy volunteers in order to determine parameters that will generate optimal signal quality. A second objective of the study was to determine reliability and validity of performing volumetric assessments. A replica of the adult human olfactory bulb and tract was constructed and three-dimensional, high-resolution imaging was performed. Our findings indicate that volumetric assessment of human olfactory structures can be achieved with high field MRI in order to examine putative differences between patients with psychotic disorders and control subjects.

T37) Do individual differences in inhibition of return and single target discrimination ability affect visual search efficiency?

Jeffery MacLeod

We investigated whether amount of inhibition of return (IOR: a tendency for slower responses to targets presented at previously attended locations) and discrimination ability can predict individual differences in visual search efficiency. There is suggestion that IOR can act as a ‘foraging facilitator’ in visual search by working as a type of implicit memory system that discourages re-inspection of items. Also, it was suspected that search efficiency should vary with individual discrimination ability, as search difficulty can vary with the similarity between targets and distractors. Three tasks were used: a single-item discrimination task (target/non-target discrimination), a visuo-spatial attention task (to measure IOR), and a visual search and probe task (to measure search efficiency and IOR to a probe presented following search). Amount of IOR following search and discrimination ability were each found to be significantly correlated with search efficiency: increasing amount of IOR and increasing discrimination ability predicted increasing search efficiency. Findings support the proposal that IOR is a foraging facilitator, and suggest that discrimination ability is a rate-determining factor in visual search.

T38) "Doing the Locomotion" With the Rat Perifornical Hypothalamus: Who's excited about glutamate?

Frederick Li, Samuel Deurveilher, Christina Morgan, and Kazue Semba

The present study examined the role of glutamatergic inputs to the perifornical lateral hypothalamus PeFLH in the regulation of various arousal-related behaviors. Rats received a unilateral microinjection of vehicle or glutamate receptor agonists, AMPA (1 or 2mM) or NMDA (10mM), and were placed into an open field for a 90-min video recording. Rats were then perfused and processed for double-label immunohistochemistry for c-Fos, a marker for neuronal activation, and orexin, a neuronal phenotype exclusive to this region. As compared to vehicle, AMPA and NMDA injections increased the total distance traveled within 90 min. The number of cells immunoreactive for both c-Fos and orexin increased at the injection site. These results suggest that glutamatergic inputs to orexin neurons in the PeFLH may play an important role in locomotor activity.

T39) Neuronal or glial origin for the electroretinogram (ERG) in insect eyes?

Steve Shaw

An insect's CNS is isolated from its circulation by a blood-brain barrier, derived from which is a blood-eye barrier (BEB). Because visual neurons cross this BEB, their electrical activity should generate responses across it. This is the conventional explanation for the unusually large extracellular waves recorded in the eye as the electroretinogram (ERG). The ON- and OFF-responses of the ERG are supposed to come from monopolar neurons directly postsynaptic to the photoreceptors, in the first synaptic zone, the lamina. The ERG is employed widely to gauge whether the first synapse is working normally, following genetic perturbations aimed at dissecting the molecular mechanisms underlying cycling of the photoreceptors' neurotransmitter, histamine.

Do monopolar neurons really generate the ERG transients? Simultaneous recordings of the ERG and intracellular recordings from visual neurons suggest not. A high resistance barrier proximal to the lamina (likely the marginal glial cells) appears to generate the ERG transients. Monopolar neurons cross this glial zone so their responses should also contribute to the ERG, and candidate wavelets exist. Their small size indicates that the ERG currents from the glia are much larger than those coming from the neurons. These glial cells are obviously light-responsive: the same histamine release may activate both glia and monopolar neurons. At least for those interested in *pre*synaptic mechanisms, ERG transients from glia merely substitute a glial origin for a response so far assumed to be neuronal.

T40) Using illusory line motion to explore attentional capture

Yoko Ishigami, Ray Klein and John Christie

Uninformative peripheral cues (figure-8 markers) and targets (an 8 changing to a 2 or 5) were combined with illusory line motion (ILM) probes to explore attentional capture. Markers were presented above, below, left and right of fixation. Any marker could brighten but for a particular participant targets only appeared on the horizontal or vertical axes. Occasionally, a diagonal line was presented connecting two of the 8s. Such lines are seen to be drawn away from a cue when the cue is adjacent to the line. We used ILM and digit identification performance to measure how endogenous attention (allocated to the 8s that could change to targets) modulates the cue,s signal strength. Digit identification reaction time and accuracy revealed that attention was captured by exogenous cues even when they were presented at task irrelevant locations. Whereas the reported magnitude of ILM away from an adjacent cue was not significantly affected by endogenous attentional control settings, ILM was weakly experienced in lines not adjacent to a cue in a manner suggesting endogenous generation of ILM.

T41) Opposite representations of object mass by the perceptual and sensorimotor systems in the size-weight illusion

Mathew S. Grandy and David A. Westwood

Goodale and Milner's (1992) duplex vision model suggests that separate visual pathways underlie object perception and the control of object-directed actions. Virtually all the research behind this model is based on tasks for which (1) vision is the primary or exclusive sensory input, and (2) grasping is the motor output. Recently, Flanagan & Beltzner (2000) showed that perceptual and motor responses to the size-weight illusion (in which a smaller object is perceived to weigh more than a larger object of identical mass) can also be dissociated. Using faulty logic, they claimed to have shown a dissociation between the effect of the SWI on perceptual reports of object mass and on the scaling of lifting forces used to pick up the object. We show in the present study using superior methods that the sensorimotor and perceptual systems clearly generate separate (and indeed opposite) representations of object mass in the SWI over repeated lifting episodes. This finding has clear and important implications for understanding multi-sensory integration for perception and action.

T42) The Affect of Short Musical Phrases with Different Melodic Contours

Joshua Salmon and Bradley Frankland

The affect (emotional content) induced by music has been assessed in longer pieces of music, mainly by reference to constructs such as modality, pitch height, tempo, and tonality. Specifically, major keys sound happy, minor keys sad, higher keys (pitches) more uplifting, faster tempos more exciting, and tonal pieces more pleasant to listen to. Note that terms like modality tempo and tonality refer to broad qualities of music leading to broad affective experience. Less is known about the affective quality of melodic contour, possibly because it is so hard to define. We propose to explore the relationship between induced affect and the attributes of the melodic contour on a note-by-note basis. Using short (8-14 notes) simple phrases of different melodic contours, the primary goal is to tease out the specific aspects of contour that lead to affective responses to music. The second goal is to create computational models of structures that lead to induced affect.

POSTERS

P1) Where did it go? Developmental Changes in Preschoolers' Search Strategies as a Function of the Type of Symbol Used for Searching

Matthew Murphy & Sophie Jacques

The current study assessed the effects of iconicity on the flexible use of cognition in young children using model-room search tasks. Three- and four-year-olds were tested using two model conditions that varied in the degree to which items in one room matched items in the other room (iconic vs. noniconic). Two types of trials that differed on whether they required the use of cognitive flexibility were also included (control vs. flexible). Results showed an effect of iconicity on both trial types, but this effect was mediated by children's age and whether they were searching in a corresponding room or in the original model suggesting a developmental change in children's efficient use of different kinds of symbols.

P2) Effect of Stimulant Medication on Daily Rhythms in Children with Attention-Deficit/Hyperactivity Disorder

Penny Corkum and Alex Dumaresq

Much research has been devoted to the study of sleep in children with ADHD. Stimulant medication, the most common form of treatment for ADHD, has been shown to negatively effect sleep. To date, no research has been done investigating how stimulant medication affects 24-hour cycles. The present study tested the hypothesis that stimulant medication negatively effects the rhythmicity of an individual,s 24-hour cycle. Four children newly diagnosed with ADHD-C and who were participating in a blinded medication trial were recruited. The medication trial consisted of four 7-day conditions: baseline, Placebo control, low dose, and medium dose MPH. Activity levels over the medication trial were recorded using actigraphy. Increased night-time activity and dampened circadian amplitude were observed in all participants during MPH conditions. These findings are discussed in terms of their implications for clinical practice and future research.

P3) Thinking vs. Doing: Effects of Active Participation and Inference on Children's Performance on the Representational Change Task

Amélie Doucet & Sophie Jacques

Research suggests that allowing active participation and decreasing inferential requirements may help preschoolers succeed on the representational-change task earlier than they typically do. Forty-eight 3-year-olds participated in one of three variations of an unexpected-contents task. In a control condition, children's initial belief about the contents of a box had to be inferred; in a see condition, children visually confirmed their initial belief; and in an active condition, children actively participated in creating their initial belief. Children also participated in a false-belief task and on separate measures assessing participation memory and inferential ability. Three-year-olds' performance did not differ across conditions on the representational-change task. In addition, there was no relation between performance on the representational-change task and any of the additional measures. Results do not support the idea that allowing participation or reducing inferential requirements facilitate representational change performance in 3-year-olds.

P4) Genuine, Suppressed, and Faked Facial Expressions of Pain in Children

Anne-Claire Larochette, Christine T. Chambers, and Kenneth D. Craig

Genuine, suppressed, and faked facial expressions of pain were examined in six boys and four girls aged 9 to 12 years old. Children submersed their hands in cold and warm water, and were given different instructions about what to show on their faces. Ten degree Celsius cold water was used for the genuine and suppressed conditions, and warm 30 degree Celsius water was used for the faked condition. The Facial Action Coding System was used to code the facial expressions in each condition, as well as a baseline condition. Faked expressions of pain in children were found to show more frequent facial actions typical of a genuine pain expression, making the faked expression distinct from the genuine expression. Children were able to successfully suppress their pain and mimic the baseline expression. The results indicate that children are capable of controlling their facial expressions of pain when instructed to do so, but may be better able to hide their pain than fake it.

P5) Statisticians in the Making? The Role of Statistical Learning in Preschoolers' Categorization Tendencies

Allison MacFadyen and Sophie Jacques

Despite the findings that statistical learning may be fundamental to cognitive development in infancy, research on statistical learning during the preschool years is still lacking. The current study was designed to assess 3- to-5-year-olds' use of statistical learning in categorization. It was proposed that children passively exposed to the statistical variability of different visual features of novel categories would exhibit higher categorization accuracy during subsequent categorization tasks compared to children not exposed to this information. The role of language was also investigated in the passive pre-exposure phase using category exemplar labels, and in the categorization phase itself using a naming task. Based on previous findings (Alexander & Enns, 1988), it was expected that younger children would show an increase in categorization accuracy in the naming task compared to a sorting task. Results were interpreted in terms of the strategies available to children in solving the categorization tasks. Although somewhat limited, the findings in this exploratory study provide a promising base for future research.

P6) Relative Attentional Load in an Item-Method Directed Forgetting Task

Jonathan M. Fawcett & Tracy L. Taylor

Reaction time (RT) was measured in response to visual probes embedded within a standard item-method directed forgetting task. Study words were presented sequentially, each followed 2000 ms later by a cue to remember (R) or forget (F). RT probes were presented at stimulus onset asynchronies (SOAs) of 400, 800 or 1600 ms in relation to the study word or memory cue. A recognition task was administered to measure retention of R-cued and F-cued words. Measured relative to post-word baselines, post-F cue RTs were longer than post-R cue RTs for SOAs of 400 ms and 800 ms. Recognition data indicated a significant directed forgetting effect favouring memory of R-cued over F-cued words. Results supported the model of item-method directed forgetting proposed by Zacks, Radvansky and Hasher (1996) suggesting that forgetting is accomplished by the inhibition of F-cued words. Further research is required to better understand the potential role of inhibition within this paradigm.

P7) Early Postnatal Pain Results in Test Apparatus Anxiety

Autumn P. Mochinski and Heather M. Schellinck

Pain during development is believed to permanently alter neurobiology and behaviour in both humans and rodents. Behavioural studies in animal models have primarily focused upon measuring the effects of early pain on subsequent anxiety in standard tests such as the elevated plus maze and the open field. The results have not been consistent either within or between labs and appear to depend on such variables as the type of pain, the age of the animals when they experienced the pain, the age when tested for anxiety and the type of apparatus in which they were tested. In the current study, CD-1 mice received injections of either 1% formalin, physiological saline or a sham treatment on Post Natal Days (PND) 3-5 and were exposed to a rat as a predator stressor on PND 30. Formalin injected animals scored significantly differently than controls on several measures of anxiety during habituation to the test apparatus but not during the test phase when the rat was present. These results would suggest that being in the test apparatus itself created such anxiety in the formalin injected animals that their levels of anxiety did not increase in the predator-stress situation.

P8) Selective cholinergic basal forebrain lesions using anti-murine-p75-saporin (mu p75-saporin): a mouse model of Alzheimer's disease?

Nicola Hoffman, Kazue Semba, Jennifer Stamp, Richard Brown

Degeneration of cholinergic neurons in the basal forebrain (BF) is a prominent characteristic of Alzheimer's disease (AD). One current animal model developed to mimic BF cholinergic neuron atrophy observed in human AD is via injecting neurotoxins into the mouse brain. Our long-term research aims are to selectively lesion mouse BF cholinergic neurons and then examine the cognitive and other functional deficits associated with this cell loss. Mu p75-saporin is an immunotoxin that has been shown to selectively destroy cholinergic neurons in the medial septum (MS), but demonstrates concurrent cell loss in cerebellar Purkinje cells. In the present study we examined the efficacy of the recently developed, presumably improved mu p75-saporin in selectively lesioning BF cholinergic neurons in the mouse brain. One of three doses (1.0, 1.5, or 2.0 μg/μl) of the immunotoxin or vehicle was infused bilaterally into the lateral ventricles of male C57BL/6J mice. Two weeks following injections, mice were intra-cardially perfused, the brains removed and sectioned for immunohistochemistry to visualize cholinergic and a subtype of GABAergic BF neurons, as well as acetylcholinesterase (AChE) histochemistry. Quantitative analysis revealed a reduction of cholinergic neurons in the MS/VDB following intracerebroventricular injections of mu p75-saporin, but only a trend for a decrease of cholinergic neurons in the nBM. The density of AChE-postive fibers in the hippocampus and somatosensory cortex was not significantly altered following neurotoxin injection, although there was a trend for a decrease in the hippocampus. No loss of noncholinergic neurons in the MS or nBM was apparent, nor was there a loss of noncholinergic cerebellar Purkinje cells. The results indicate that the most recently developed mu p75-saporin has an increased selectivity to cholinergic MS neurons, compared with the previous formula, and may be useful as a selective cholinergic immunotoxin for mice. Mu 75-saporin may be a useful tool to create an animal model of AD and expand research concerning the progression and neuropathology of the disease.